Gaussian product blocking: the tree-level coupling never moves

For the free lattice gauge field in D dimensions with heat time t, one isotropic product-blocking step $a \mapsto 2a$ gives

$$C'(K) = 2^{4-D} t \, [P_{\mathrm{cont}}(\hat{K}) + O(K)]$$

on exact 2-forms, with $O(K^2)$ in plaquette-centred conventions for the nearest-plaquette input (Theorem G1). The coefficient is exactly $\lambda_D(2a)^{4-D}$. The statement holds for positive input covariances supported on exact forms, bounded near nonzero aliases, with the same small-momentum expansion. It iterates at every finite depth. The four directional integrations compose by Fubini, preserving the tree-level coupling in the geometric heat-time normalization. In $1+3$, t stays fixed.

Consequence for cell 4. In (15) and the many-steps remark of the four-dimensional note, the Gaussian zero-momentum normalization introduces no matching shift. The one-loop endpoint term depends on the full shape and its non-abelian background completion, including generated cubic and quartic vertices. The average rate needs $c_n = o(n)$; logarithmic growth of c_n suffices. Growth of shape parameters needs a separate estimate relating it to c_n.

Caution. Sharp product blocking retains fluxes through sharp surfaces. Their ultraviolet boundary fluctuations obstruct a finite nondegenerate covariance fixed point (composition note, Theorem 2). Bell–Wilson 1975 (metadata and abstract) supplies the Gaussian fixed-point framework. Bietenholz–Wiese 1996 (§3, equations (3.1)–(3.6), passage) uses spatially averaged gauge variables even with an exact delta constraint. These are distinct choices of blocking map.

The estimates are internal written checks; constants may depend on finite iteration depth. No independent peer review is claimed.

1. Setting

Take $G = \mathbb{R}$ (colour components decouple) on a periodic lattice of spacing a, link variables A, fluxes $\phi = dA$ and weight $\prod_p e^{-\phi_p^2/(2t)}$. After gauge reduction the flux covariance is $tP(k)$, the orthogonal projector onto exact 2-forms for $k \neq 0$. The harmonic sector is set aside. Here zero-momentum coefficient means the limit through nonzero momenta after the bulk limit; one may also adjoin a constant harmonic background with its Maxwell action.

Product blocking sums the two fine links along each coarse link. Stokes gives

$$(B\phi)_{\mu\nu}(X) = \sum_{a,b\in\{0,1\}} \phi_{\mu\nu}(2X + ae_\mu + be_\nu).$$

This composes the Gaussian pushforwards of Proposition 1 of the series/parallel note.

2. Theorem G1 and proof

Alias sum. With unitary Fourier transforms, $k_n = K/2 + \pi n$, $n \in \{0,1\}^D$, and

$$\begin{aligned}
(B\phi)^{\wedge}_{\mu\nu}(K) &= 2^{-D/2} \sum_n f_{\mu\nu}(k_n) \hat{\phi}_{\mu\nu}(k_n), \\
f_{\mu\nu}(k) &= (1 + e^{ik_\mu})(1 + e^{ik_\nu}), \\
C'(K) &= 2^{-D} t \sum_n F(k_n) P(k_n) F(k_n)^*, \quad F = \operatorname{diag} f.
\end{aligned}$$

The term $n = 0$. Here $|f_{\mu\nu}(K/2)|^2 = 16\cos^2(K_\mu/4)\cos^2(K_\nu/4)$. The lattice projector is $P_{\mathrm{cont}}(\hat{K}) + O(K)$. With the phases at plaquette centres, the curl uses $2\sin(k_j/2)$, the block factors become $4\cos(K_\mu/4)\cos(K_\nu/4)$, and the error is $O(K^2)$. The contribution is therefore $2^{4-D}t[P_{\mathrm{cont}}(\hat{K}) + O(K^2)]$ in this convention.

Staggered terms. If either n_μ or n_ν is 1, the corresponding factor in $f_{\mu\nu}$ is $O(K)$. If both are zero, choose $n_\lambda = 1$ outside the pair. On the exact subspace, closedness gives

$$d_\lambda \phi_{\mu\nu} = d_\nu \phi_{\mu\lambda} - d_\mu \phi_{\nu\lambda}, \qquad d_j = e^{ik_j} - 1.$$

For $|K| \le 1$, its component functional has norm at most $|K|/(4\cos(1/4)) \le |K|/2$. Equivalently, applying the inequality to $Pe_{\mu\nu}$ and using $\langle e_{\mu\nu}, Pe_{\mu\nu}\rangle = \|Pe_{\mu\nu}\|^2$ gives $\|Pe_{\mu\nu}\| \le |K|/2$. When both form factors are unsuppressed, use this bound twice in $P_{\mu\nu,\rho\sigma} = \langle Pe_{\mu\nu}, Pe_{\rho\sigma}\rangle$; each pair may choose its own staggered witness. When a form factor is suppressed it supplies the corresponding power. Every contribution is $O(K^2)$.

More explicitly, put $m = \binom{D}{2}$. The columns give $\|PF^*\| \le 2\sqrt{m}|K|$. For a general covariance $C = PCP$ with $\|C\| \le M$ near the nonzero aliases, their total contribution has operator norm at most

$$2^{-D}(2^D - 1)4mM|K|^2.$$

Iteration. Assume in addition $C(k) = t[P_{\mathrm{cont}}(\hat{k}) + O(k)]$ at the origin (or the centred $O(k^2)$ version for that stronger conclusion). The preceding bound handles the nonzero aliases, and the small-momentum hypothesis handles $n = 0$. The output retains exact support by Stokes and is bounded away from the origin. Hence induction multiplies t by 2^{4-D} at each finite step. The nearest-plaquette input and all its finite iterates have the centred $O(k^2)$ property. The remainder constants need no bound uniform in iteration depth for this conclusion.

Fubini composes the directional integrations. Their constant-flux coefficient agrees with the reference geometric heat times; the matrix defect and the full composed kernel are given in Theorem 1 of the composition note. □

3. What the theorem leaves to the interacting theory

The abelian Gaussian has no loop correction to the coupling. In a non-abelian background the determinant depends on its background Hessian, so G1 fixes only the tree-level normalization. The full momentum dependence and the background vertices need separate bounds. The composition note gives an exact determinant continuity criterion and states the remaining vertex and infrared estimates explicitly.

4. Consequence for STATE

G1 closes the Gaussian zero-momentum part at every finite depth. Follow the composition note for sharp-blocking shape growth and the one-loop matching criterion; the average-rate condition is $c_n = o(n)$.